

Skill week-1

Creating Database, Table and Inserting Data Into Table.

SQL File 3* x

```
1 • create DATABASE sweek1;
2
3 • USE sweek1;
4
5 • CREATE TABLE student_marks1 (
6     roll_no INT PRIMARY KEY,
7     name VARCHAR(50),
8     subject VARCHAR(50),
9     marks DECIMAL(5,2)
10 );
11
12 • INSERT INTO student_marks1 (roll_no, name, subject, marks) VALUES
13 (1, 'Ravi', 'Math', 85.50),
14 (2, 'Sita', 'Math', 92.75),
15 (3, 'Anil', 'Math', 78.40),
16 (4, 'Priya', 'Math', 88.90),
17 (5, 'Vijay', 'Math', 80.25);
18
```

Output

#	Time	Action	Message
1	10:58:37	create DATABASE sweek1	Error Code: 1007. Can't create database 'sweek1'; database exists
2	10:58:44	USE sweek1	0 row(s) affected
3	10:58:49	CREATE TABLE student_marks1 (roll_no INT PRIMARY KEY, name VARCHAR(50), subject VARCHAR(50), ...	0 row(s) affected
4	10:59:17	INSERT INTO student_marks (roll_no, name, subject, marks) VALUES (1, 'Ravi', 'Math', 85.50), (2, 'Sita', 'Math', 92.7...	Error Code: 1064. You have an error in your SQL syntax; check the manual that correspon
5	10:59:39	INSERT INTO student_marks (roll_no, name, subject, marks) VALUES (1, 'Ravi', 'Math', 85.50), (2, 'Sita', 'Math', 92.7...	Error Code: 1146. Table 'sweek1.student_marks' doesn't exist
6	10:59:52	INSERT INTO student_marks1 (roll_no, name, subject, marks) VALUES (1, 'Ravi', 'Math', 85.50), (2, 'Sita', 'Math', 92...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0

SQL File 3* x

```
3 • SELECT * FROM student_marks1;
4
```

Result Grid

roll_no	name	subject	marks
1	Ravi	Math	85.50
2	Sita	Math	92.75
3	Anil	Math	78.40
4	Priya	Math	88.90
5	Vijay	Math	80.25
NULL	NULL	NULL	NULL

dent_marks1 1 x

Output

#	Time	Action
1	11:20:04	use sweek1
2	11:20:21	SELECT * FROM student_marks1 LIMIT 0, 1000

```
2
3 • SELECT * FROM student_marks1;
4
5 • SELECT COUNT(*) AS total_students FROM student_marks1;
6 • SELECT SUM(marks) FROM student_marks1;
7 • SELECT AVG(marks) FROM student_marks1;
8 • SELECT MAX(marks) FROM student_marks1;
9 • SELECT MIN(marks) FROM student_marks1;
10
```

Result 2 x

Output

#	Time	Action
1	11:20:04	use sweek1
2	11:20:21	SELECT * FROM student_marks1 LIMIT 0, 1000
3	11:32:42	SELECT COUNT(*) AS total_students FROM student_marks1 LIMIT 0, 1000

```

5 • SELECT COUNT(*) AS total_students FROM student_marks1;
6 • SELECT SUM(marks) FROM student_marks1;
7 • SELECT AVG(marks) FROM student_marks1;
8 • SELECT MAX(marks) FROM student_marks1;
9 • SELECT MIN(marks) FROM student_marks1;
10

```

Result Grid

SUM(marks)
425.80

Result 3 x

Output

Action Output

#	Time	Action
1	11:20:04	use sweek1
2	11:20:21	SELECT * FROM student_marks1 LIMIT 0, 1000
3	11:32:42	SELECT COUNT(*) AS total_students FROM student_marks1 LIMIT 0, 1000
4	11:33:36	SELECT SUM(marks) FROM student_marks1 LIMIT 0, 1000

```

5 • SELECT COUNT(*) AS total_students FROM student_marks1;
6 • SELECT SUM(marks) FROM student_marks1;
7 • SELECT AVG(marks) FROM student_marks1;
8 • SELECT MAX(marks) FROM student_marks1;
9 • SELECT MIN(marks) FROM student_marks1;
10

```

Result Grid

AVG(marks)
85.160000

Result 4 x

Output

Action Output

#	Time	Action
1	11:20:04	use sweek1
2	11:20:21	SELECT * FROM student_marks1 LIMIT 0, 1000
3	11:32:42	SELECT COUNT(*) AS total_students FROM student_marks1 LIMIT 0, 1000
4	11:33:36	SELECT SUM(marks) FROM student_marks1 LIMIT 0, 1000
5	11:34:04	SELECT AVG(marks) FROM student_marks1 LIMIT 0, 1000

```

5 • SELECT COUNT(*) AS total_students FROM student_marks1;
6 • SELECT SUM(marks) FROM student_marks1;
7 • SELECT AVG(marks) FROM student_marks1;
8 • SELECT MAX(marks) FROM student_marks1;
9 • SELECT MIN(marks) FROM student_marks1;
10

```

Result Grid

MAX(marks)
92.75

Result 5 x

Output

Action Output

#	Time	Action
1	11:20:04	use sweek1
2	11:20:21	SELECT * FROM student_marks1 LIMIT 0, 1000
3	11:32:42	SELECT COUNT(*) AS total_students FROM student_marks1 LIMIT 0, 1000
4	11:33:36	SELECT SUM(marks) FROM student_marks1 LIMIT 0, 1000
5	11:34:04	SELECT AVG(marks) FROM student_marks1 LIMIT 0, 1000
6	11:34:27	SELECT MAX(marks) FROM student_marks1 LIMIT 0, 1000

```

5 • SELECT COUNT(*) AS total_students FROM student_marks1;
6 • SELECT SUM(marks) FROM student_marks1;
7 • SELECT AVG(marks) FROM student_marks1;
8 • SELECT MAX(marks) FROM student_marks1;
9 • SELECT MIN(marks) FROM student_marks1;
10

```

Result Grid

MIN(marks)
78.40

Result 6 x

Output

Action Output

#	Time	Action
1	11:20:04	use sweek1
2	11:20:21	SELECT * FROM student_marks1 LIMIT 0, 1000
3	11:32:42	SELECT COUNT(*) AS total_students FROM student_marks1 LIMIT 0, 1000
4	11:33:36	SELECT SUM(marks) FROM student_marks1 LIMIT 0, 1000
5	11:34:04	SELECT AVG(marks) FROM student_marks1 LIMIT 0, 1000
6	11:34:27	SELECT MAX(marks) FROM student_marks1 LIMIT 0, 1000
7	11:34:45	SELECT MIN(marks) FROM student_marks1 LIMIT 0, 1000

Filtering with WHERE

The screenshot displays a SQL editor with several queries using the WHERE clause. Below the editor, a 'Result Grid' shows the output of the selected query. The 'Output' pane at the bottom provides a log of executed actions and their results.

SQL Queries:

```
8 • SELECT MAX(marks) FROM student_marks1;
9
10 • SELECT * FROM student_marks1 WHERE marks > 85;
11 • SELECT * FROM student_marks1 WHERE marks >= 90;
12 • SELECT * FROM student_marks1 WHERE marks < 80;
13 • SELECT * FROM student_marks1 WHERE marks BETWEEN 80 AND 90;
14 • SELECT * FROM student_marks1 WHERE name LIKE 'P%';
15 • SELECT * FROM student_marks1 WHERE name IN ('Ravi', 'Sita', 'Vijay');
16 • SELECT * FROM student_marks1 WHERE marks > 85 AND (subject = 'Math' OR name LIKE 'P%');
17
```

Result Grid:

roll_no	name	subject	marks
1	Ravi	Math	85.50
2	Sita	Math	92.75
4	Priya	Math	88.90
NULL	NULL	NULL	NULL

Output Log:

#	Time	Action	Message
8	11:36:49	SELECT * FROM student_marks1 WHERE marks > 85 LIMIT 0, 1000	3 row(s) returned
9	11:36:57	SELECT * FROM student_marks1 WHERE marks >= 90 LIMIT 0, 1000	1 row(s) returned
10	11:37:00	SELECT * FROM student_marks1 WHERE marks < 80 LIMIT 0, 1000	1 row(s) returned
11	11:37:05	SELECT * FROM student_marks1 WHERE marks BETWEEN 80 AND 90 LIMIT 0, 1000	3 row(s) returned
12	11:37:10	SELECT * FROM student_marks1 WHERE name LIKE 'P%' LIMIT 0, 1000	1 row(s) returned
13	11:37:14	SELECT * FROM student_marks1 WHERE name IN ('Ravi', 'Sita', 'Vijay') LIMIT 0, 1000	3 row(s) returned
14	11:37:17	SELECT * FROM student_marks1 WHERE marks > 85 AND (subject = 'Math' OR name LIKE 'P%') LIMIT 0, 1000	3 row(s) returned

Updating Queries

The screenshot shows a SQL editor with an UPDATE query. The 'Output' pane at the bottom shows the execution log.

SQL Query:

```
1 • use sweek1;
2
3 • UPDATE student_marks1 SET marks = 90.00 WHERE roll_no = 3;
```

Output Log:

#	Time	Action
1	14:33:19	use sweek1
2	14:33:22	UPDATE student_marks1 SET marks = 90.00 WHERE roll_no = 3

Deleting Queries

```
1 • use sweek1;
2
3 • UPDATE student_marks1 SET marks = 90.00 WHERE roll_no = 3;
4
5 • DELETE FROM student_marks1 WHERE roll_no = 5;
6 • DELETE FROM student_marks1 WHERE marks < 75;
7
```

Output

Action Output

#	Time	Action
✓ 1	14:33:19	use sweek1
✓ 2	14:33:22	UPDATE student_marks1 SET marks = 90.00 WHERE roll_no = 3
✗ 3	14:41:35	DELETE FROM student_marks WHERE roll_no = 5
✓ 4	14:41:51	DELETE FROM student_marks1 WHERE roll_no = 5
✗ 5	14:41:54	DELETE FROM student_marks1 WHERE marks < 75

Sorting

```
7
8 • SELECT * FROM student_marks1 ORDER BY marks ASC;
9 • SELECT * FROM student_marks1 ORDER BY marks DESC;
10 • SELECT * FROM student_marks1 ORDER BY name ASC;
11 • SELECT * FROM student_marks1 ORDER BY name DESC;
12
```

Result Grid

roll_no	name	subject	marks
1	Ravi	Math	85.50
4	Priya	Math	88.90
3	Anil	Math	90.00
2	Sita	Math	92.75
*	NULL	NULL	NULL

student_marks11 x

Output

Action Output

#	Time	Action
✓ 1	14:33:19	use sweek1
✓ 2	14:33:22	UPDATE student_marks1 SET marks = 90.00 WHERE roll_no = 3
✗ 3	14:41:35	DELETE FROM student_marks WHERE roll_no = 5
✓ 4	14:41:51	DELETE FROM student_marks1 WHERE roll_no = 5
✗ 5	14:41:54	DELETE FROM student_marks1 WHERE marks < 75
✓ 6	14:44:25	SELECT * FROM student_marks1 ORDER BY marks ASC LIMIT 0, 1000
✓ 7	14:45:15	SELECT * FROM student_marks1 ORDER BY marks DESC LIMIT 0, 1000

```
8 • SELECT * FROM student_marks1 ORDER BY marks ASC;
9 • SELECT * FROM student_marks1 ORDER BY marks DESC;
10 • SELECT * FROM student_marks1 ORDER BY name ASC;
11 • SELECT * FROM student_marks1 ORDER BY name DESC;
12
```

Result Grid

roll_no	name	subject	marks
2	Sita	Math	92.75
3	Anil	Math	90.00
4	Priya	Math	88.90
1	Ravi	Math	85.50
*	NULL	NULL	NULL

student_marks12 x

Output

Action Output

#	Time	Action
✓ 1	14:33:19	use sweek1
✓ 2	14:33:22	UPDATE student_marks1 SET marks = 90.00 WHERE roll_no = 3
✗ 3	14:41:35	DELETE FROM student_marks WHERE roll_no = 5
✓ 4	14:41:51	DELETE FROM student_marks1 WHERE roll_no = 5
✗ 5	14:41:54	DELETE FROM student_marks1 WHERE marks < 75
✓ 6	14:44:25	SELECT * FROM student_marks1 ORDER BY marks ASC LIMIT 0, 1000
✓ 7	14:45:15	SELECT * FROM student_marks1 ORDER BY marks DESC LIMIT 0, 1000

```

8 • SELECT * FROM student_marks1 ORDER BY marks ASC;
9 • SELECT * FROM student_marks1 ORDER BY marks DESC;
10 • SELECT * FROM student_marks1 ORDER BY name ASC;
11 • SELECT * FROM student_marks1 ORDER BY name DESC;
12

```

Result Grid

roll_no	name	subject	marks
2	Sita	Math	92.75
1	Ravi	Math	85.50
4	Priya	Math	88.90
3	Anil	Math	90.00
NULL	NULL	NULL	NULL

student_marks1 4 ×

Output

Action Output

#	Time	Action
3	14:41:35	DELETE FROM student_marks WHERE roll_no = 5
4	14:41:51	DELETE FROM student_marks1 WHERE roll_no = 5
5	14:41:54	DELETE FROM student_marks1 WHERE marks < 75
6	14:44:25	SELECT * FROM student_marks1 ORDER BY marks ASC LIMIT 0, 1000
7	14:45:15	SELECT * FROM student_marks1 ORDER BY marks DESC LIMIT 0, 1000
8	14:47:45	SELECT * FROM student_marks1 ORDER BY name ASC LIMIT 0, 1000
9	14:47:52	SELECT * FROM student_marks1 ORDER BY name DESC LIMIT 0, 1000

Group By

```

13 • SELECT subject, SUM(marks) AS total_marks FROM student_marks1 GROUP BY subject;
14 • SELECT subject, AVG(marks) AS avg_marks FROM student_marks1 GROUP BY subject;
15 • SELECT subject, COUNT(*) AS num_students FROM student_marks1 GROUP BY subject;
16

```

Result Grid

subject	num_students
Math	4

Result 7 ×

Output

Action Output

#	Time	Action
6	14:44:25	SELECT * FROM student_marks1 ORDER BY marks ASC LIMIT 0, 1000
7	14:45:15	SELECT * FROM student_marks1 ORDER BY marks DESC LIMIT 0, 1000
8	14:47:45	SELECT * FROM student_marks1 ORDER BY name ASC LIMIT 0, 1000
9	14:47:52	SELECT * FROM student_marks1 ORDER BY name DESC LIMIT 0, 1000
10	14:48:56	SELECT subject, SUM(marks) AS total_marks FROM student_marks1 GROUP BY subject LIMIT 0, 1000
11	14:49:01	SELECT subject, AVG(marks) AS avg_marks FROM student_marks1 GROUP BY subject LIMIT 0, 1000
12	14:49:05	SELECT subject, COUNT(*) AS num_students FROM student_marks1 GROUP BY subject LIMIT 0, 1000

Having

17 • SELECT subject, AVG(marks) AS avg_marks

18 FROM student_marks1

19 GROUP BY subject

20 HAVING AVG(marks) > 90;

21

22 • SELECT subject, COUNT(*) AS num_students

23 FROM student_marks1

24 GROUP BY subject

25 HAVING COUNT(*) > 1;

26

Result Grid

subject

avg_marks

Result 8

Output

Action Output

#	Time	Action
8	14:47:45	SELECT * FROM student_marks1 ORDER BY name ASC LIMIT 0, 1000
9	14:47:52	SELECT * FROM student_marks1 ORDER BY name DESC LIMIT 0, 1000
10	14:48:56	SELECT subject, SUM(marks) AS total_marks FROM student_marks1 GROUP BY subject LIMIT 0, 1000
11	14:49:01	SELECT subject, AVG(marks) AS avg_marks FROM student_marks1 GROUP BY subject LIMIT 0, 1000
12	14:49:05	SELECT subject, COUNT(*) AS num_students FROM student_marks1 GROUP BY subject LIMIT 0, 1000
13	14:50:08	SELECT subject, AVG(marks) AS avg_marks FROM student_marks1 GROUP BY subject HAVING AVG(marks) > 90...
14	14:50:22	SELECT subject, AVG(marks) AS avg_marks FROM student_marks1 GROUP BY subject HAVING AVG(marks) > 9...

17 • SELECT subject, AVG(marks) AS avg_marks

18 FROM student_marks1

19 GROUP BY subject

20 HAVING AVG(marks) > 90;

21

22 • SELECT subject, COUNT(*) AS num_students

23 FROM student_marks1

24 GROUP BY subject

25 HAVING COUNT(*) > 1;

26

Result Grid

subject

num_students

Math

4

Result 9

Output

Action Output

#	Time	Action
9	14:47:52	SELECT * FROM student_marks1 ORDER BY name DESC LIMIT 0, 1000
10	14:48:56	SELECT subject, SUM(marks) AS total_marks FROM student_marks1 GROUP BY subject LIMIT 0, 1000
11	14:49:01	SELECT subject, AVG(marks) AS avg_marks FROM student_marks1 GROUP BY subject LIMIT 0, 1000